

CRYPTOLOGY ASSIGNMENT - 3

① Determine the multiplicative inverse mod 26 for each of the following integers:

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

$3^{-1} \text{ mod } 26$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1) \quad \text{--- (1)}$$

$$2 = 26 + 3(-8) \quad \text{--- (2)}$$

$$1 = 3 + 2(-1)$$

$$1 = 3 + [26 + 3(-8)](-1)$$

$$1 = 3 + 26(-1) + 3(8)$$

$$3(9) + 26(-1) = 1$$

$$\underline{\underline{3^{-1} = 9}}$$

$5^{-1} \text{ mod } 26$

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$\underline{\underline{5^{-1} = 21}}$$

7^{-1}

$$26 = 7(3) + 5$$

$$5 = 26 + 7(-3)$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$2 = 7 + 5(-1)$$

$$5 + 2(-2) = 1$$

$$5 + 2[7 + 5(-1)](-2) = 1$$

$$5 + 7(-2) + 5(2) = 1$$

$$26 + 7(3) + 7(-2) + [26 + 7(3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$\underline{\underline{7^{-1} = 15}}$$

9⁻¹

$$26 = 9(2) + 8$$

$$8 = 26 + 9(-2)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$\underline{\underline{9^{-1} = 3}}$$

11⁻¹

$$26 = 11(2) + 4$$

$$4 = 26 + 11(-2)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + (11 + 4(-2))(-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + (26 + 11(-2))(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-9) = 1$$

$$26 - 9 = 19$$

$$\underline{\underline{11^{-1} = 19}}$$

$$15^{-1}$$

$$26 = 15(1) + 11$$

$$15 = 11(1) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + (11 + 4(-2)) - (-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + (26 + 15(-1))(-2) + (15 + 11(-1))(2) = 1$$

$$15 + 26(-2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(-2) + (26 + 15(-1))(-2) = 1$$

$$15(3) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(4) = 1$$

$$\underline{15^{-1} = 7}$$

$$17^{-1}$$

$$26 = 17(1) + 9$$

$$17 = 9(1) + 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$26 + 17(-1) + (17 + 9(-1))(-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-2) + 26 + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = 23$$

$$\underline{17^{-1} = 23}$$

$$\begin{aligned}
 19^{-1} \quad 26 &= 19(1) + 7 & 7 &= 26 + 19(-1) \\
 19 + 7(2) &= 5 & 5 &= 19 + 7(-2) \\
 7 &= 5(1) + 2 & 2 &= 7 + 5(-1) \\
 5 &= 2(2) + 1 & & \\
 5 + 2(-2) &= 1 & &
 \end{aligned}$$

$$\begin{aligned}
 19 + 7(-2) + (7 + 5(-1))(-2) &= 1 \\
 19 + 7(-2) + 7(-2) + 5(2) &= 1 \\
 19 + 7(-4) + 5(2) &= 1 \\
 19 + (26 + 19(-1))(-4) + (19 + 7(-2))(2) &= 1 \\
 19 + 26(-4) + 19(4) + 19(2) + 7(-4) &= 1 \\
 19(7) + 26(-4) + (26 + 19(-1))(-4) &= 1 \\
 19(7) + 26(-4) + 26(-4) + 19(4) &= 1 \\
 19(11) + 26(-8) &= 1
 \end{aligned}$$

$$19^{-1} = 11$$

$$\begin{aligned}
 21^{-1} \quad 26 &= 21(1) + 5 & 5 &= 26 + 21(-1) \\
 21 &= 5(4) + 1 & 1 &= 21 + 5(-4) \\
 21 + 5(-4) &= 1 & & \\
 21 + (26 + 21(-1))(-4) &= 1 \\
 21 + 26(-4) + 21(4) &= 1 \\
 26(-4) + 21(5) &= 1 \\
 21^{-1} &= 5
 \end{aligned}$$

$$23^{-1}$$

$$\begin{aligned}
 26 &= 23(1) + 3 & 3 &= 26 + 23(-1) \\
 23 &= 3(7) + 2 & 2 &= 23 + 3(-9) \\
 3 &= 2(1) + 1 & & \\
 3 + 2(-1) &= 1 & & \\
 26 + 23(-1) + (23 + 3(-9))(-1) &= 1 \\
 26 + 23(-2) + 23(-1) + (26 + 23(-1))(7) &= 1 \\
 26 + 23(-2) + 26(7) + 23(-7) &= 1 \\
 26(8) + 23(-9) &= 1 \Rightarrow 26^{-1} = 8, 23^{-1} = 19
 \end{aligned}$$

25-1

$$26 = 25(-1) + 1$$

$$26 + 25(-1) + 1$$

$$26 - 1 = 25$$

$$25 - 1 = 24$$

2) Choose any string of your choice to encrypt using the affine cipher with key value multiplier 5 and additive 8.

PLAINTEXT : ADITI

0 3 8 19 8

$$A \Rightarrow (0 \times 5 + 8) \bmod 26 \Rightarrow 8$$

$$D \Rightarrow (3 \times 5 + 8) \bmod 26 \Rightarrow 23$$

$$I \Rightarrow (8 \times 5 + 8) \bmod 26$$

$$48 \bmod 26 = 22$$

$$T \Rightarrow (19 \times 5 + 8) \bmod 26$$

$$103 \bmod 26 = 25$$

$$I \Rightarrow (8 \times 5 + 8) \bmod 26$$

$$48 \bmod 26 = 22$$

Cipher text : IXWZV

3) Encrypt 'HI' using affine cipher with $A=11$, $b=5$

$$C_2 (P \times a + b) \bmod 26$$

$$H=7 \quad I=8$$

$$C_2 (P \times 11 + 5) \bmod 26$$

$$C_2 H_2 (7 \times 11 + 5) \bmod 26$$

$$= (77 + 5) \bmod 26 = 82 \bmod 26$$

$$= 4 \Rightarrow E$$

$$C_2 I_2 (8 \times 11 + 5) \bmod 26$$

$$= (88 + 5) \bmod 26$$

$$= 93 \bmod 26$$

$$= 15 \Rightarrow F$$

$$\Rightarrow \underline{\underline{EF}}$$

4) The ciphertext UCR was encrypted using the affine function $(ax+2) \bmod 26$ find the plaintext

$$C_2 (aP + 2) \bmod 26$$

$$a=9$$

$$P = a^{-1} (C - b) \bmod 26$$

$$b=2$$

$$U=20, C=2, R=17$$

$$9^{-1} \quad 26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$\underline{\underline{9^{-1} = 3}}$$

$$8 = 26 + 9(-2)$$

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$$V = 3(20 - 2) \bmod 26 = 3(18) \bmod 26 \Rightarrow 54 \bmod 26 = 2 \Rightarrow C$$

$$C = 2$$

$$P = 3(2 - 2) \bmod 26 = 0 \Rightarrow A$$

$$R = 17$$

$$\begin{aligned} \Rightarrow P &= 3(17 - 2) \bmod 26 \\ &= 3(15) \bmod 26 \\ &= 45 \bmod 26 \Rightarrow 19 \\ 19 &\Rightarrow T \end{aligned}$$

$$= \underline{\underline{CAT}}$$

5) Decrypt the cipher text using affine cipher
K D H X J U W D E X Z Y using the key (multiplier) and 4

$$a = 3 \quad b = 4$$

$$P = a^{-1}(C - b) \bmod 26$$

K D H X J U W D E X Z Y

10 3 24 23 9 20 23 3 4 23 25 24

$$a^{-1} = 3^{-1}$$

$$2 = 26 + 3(-8)$$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$3 + (26 + 3(-8))(-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$26(-1) + 3(9) = 1$$

$$3^{-1} = 9 = a^{-1}$$

=

$$k = 9(10-4) \bmod 26$$

$$= 9(6) \bmod 26 = 2 \Rightarrow C$$

$$D = 9(3-4) \bmod 26$$

$$9(-1) \bmod 26 = 17 \Rightarrow R$$

$$Y = 9(4-4) \bmod 26 \Rightarrow 9(0) \bmod 26 \Rightarrow 24 \Rightarrow Y$$

$$X = 9(3-4) \bmod 26$$

$$9(-1) \bmod 26 = 15 \Rightarrow P$$

$$J = 9(9-4) \bmod 26 \Rightarrow 9(5) \bmod 26 = 19 \Rightarrow T$$

$$U = 9(20-4) \bmod 26$$

$$= 9(16) \bmod 26 = 14 \Rightarrow O$$

$$W = 9(22-4) \bmod 26$$

$$9(18) \bmod 26 = 16 \Rightarrow G$$

$$D = 9(3-4) \bmod 26 = 9(-1) \bmod 26 = 17 \Rightarrow R$$

$$E = 9(4-4) \bmod 26 = 9(0) \bmod 26 = 0 \Rightarrow A$$

$$X = 9(23-4) \bmod 26 = 9(19) \bmod 26 = 15 \Rightarrow P$$

$$Z = 9(25-4) \bmod 26 = 9(21) \bmod 26 = 7 \Rightarrow H$$

$$Y = 9(24-4) \bmod 26$$

$$= 9(20) \bmod 26 = 24 \Rightarrow Y$$

CRYPTOGRAPHY

6) use $k=13$ (multi) $k=24$ (even though it is not a valid key try to encrypt. Encrypt the string and what is the issue

$$C = (P \times \text{multi} + \text{add}) \bmod 26$$

$$C = (P \times a + b) \bmod 26$$

$$a = 13 \quad b = 4$$

13 is not coprime of 26
 so we cannot find inverse of 13 mod 26
 Therefore it is not a valid key for affine

for input

$$T = (8 \times 13 + 4) \bmod 26 = 108 \bmod 26 = 4 \Rightarrow E$$

$$N = (13 \times 13 + 4) \bmod 26 = 173 \bmod 26 = 17 \Rightarrow R$$

$$P = (15 \times 13 + 4) \bmod 26 = 199 \bmod 26 = 17 \Rightarrow R$$

$$U = (20 \times 13 + 4) \bmod 26 = 264 \bmod 26 = 4 \Rightarrow E$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 \Rightarrow R$$

ERROR

ALTER

$$A = (0 \times 13 + 4) \bmod 26 = 4 \bmod 26 = 4 \Rightarrow E$$

$$L = (11 \times 13 + 4) \bmod 26 = 147 \bmod 26 = 17 \Rightarrow R$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 \Rightarrow R$$

$$E = (4 \times 13 + 4) \bmod 26 = 56 \bmod 26 = 4 \Rightarrow E$$

$$R = (17 \times 13 + 4) \bmod 26 = 225 \bmod 26 = 17 \Rightarrow R$$

ERROR

Both the plaintext will produce the same ciphertext.
 The encryption is not one to one and decryption is not possible as 13 is not a valid key

7) Find the inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \text{ mod } 26$

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \text{ mod } 26$$

$$\det(K) = 3 \times 7 - 2 \times 5 = 11$$

$$11^{-1} \text{ mod } 26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 26 + 11(-2)$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(1) = 1$$

$$26 + 11(-2) + (11 + 4(-2))(-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

Adjugent of the matrix

$$\begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8) Find the inverse of the following matrix whose entries are considered module 26:

$$\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$$

$$K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \text{ mod } 26$$

$$\det(K) = 11 \times 3 - 13 \times 2 = 33 - 26 \\ = \underline{7}$$

$$26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 = 26 + 7(-3)$$

$$2 = 7 + 5(-1)$$

$$5 + 2(-2) = 1$$

$$26 + 7(-3) + 2(7 + 5(-1))(-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + (26 + 7(-3))(2) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

Adjugent is,

$$\text{adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9) Is the key valid $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$?

→ for a key to be valid determinant should be coprime and should not be equal to 0

$$\begin{aligned} \det(K) &= 2(21) - 14(13) \\ &= 42 - 182 \\ &= -140 \end{aligned}$$

$$-140 \bmod 26 = 16$$

$$\det K = 16 \bmod 26$$

16 is not coprime to 26

Therefore it will not have an inverse modulo 26

∴ The key $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$ is not valid

10) Encrypt using 2×2 hill cipher for the plaintext
SOLVED using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$3 \times 1 - 7 \times 8 = -23 = 3 \text{ which is coprime}$$

$$\text{So } \text{LVE} \quad C_2 KXP$$

$$P_2 \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 250 \end{bmatrix} \bmod 26$$

$$166 \bmod 26 = 10 = K$$

$$250 \bmod 26 = 20 = U$$

$$\begin{matrix} L \\ V \end{matrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \bmod 26$$

201 mod 26 = 19 = T

308 mod 26 = 22 = W

$$\begin{matrix} E & \begin{bmatrix} 4 \\ 3 \end{bmatrix} \\ D & \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \end{matrix}$$

$$= \begin{bmatrix} 12 & + 24 \\ 28 & + 33 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 36 \\ 61 \end{bmatrix} \text{ mod } 26$$

36 mod 26 = 10 = K

61 mod 26 = 9 = J

SOLVED $\rightarrow$ KUTWkj